SOC-in-a-BOX Implementation Guide

Lab Components

Component	Function
Ubuntu Server 22.04	Wazuh Manager, Indexer, Dashboard
Windows 10	Monitored Endpoint
Kali Linux	Attack Simulation Platform
VirtualBox	Virtualization Environment

VirtualBox VM Specifications

Wazuh Server

Setting	Value
OS	Ubuntu Server 22.04
RAM	8 GB
CPU	2 vCPU
Storage	50 GB

Windows Endpoint

Setting	Value
OS	Windows 10
RAM	4 GB
CPU	2 vCPU
Storage	50 GB

Kali Linux

Setting	Value
OS	Kali Linux
RAM	4 GB
CPU	2 vCPU
Storage	30 GB

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4. Wazuh Deployment

Installation Method

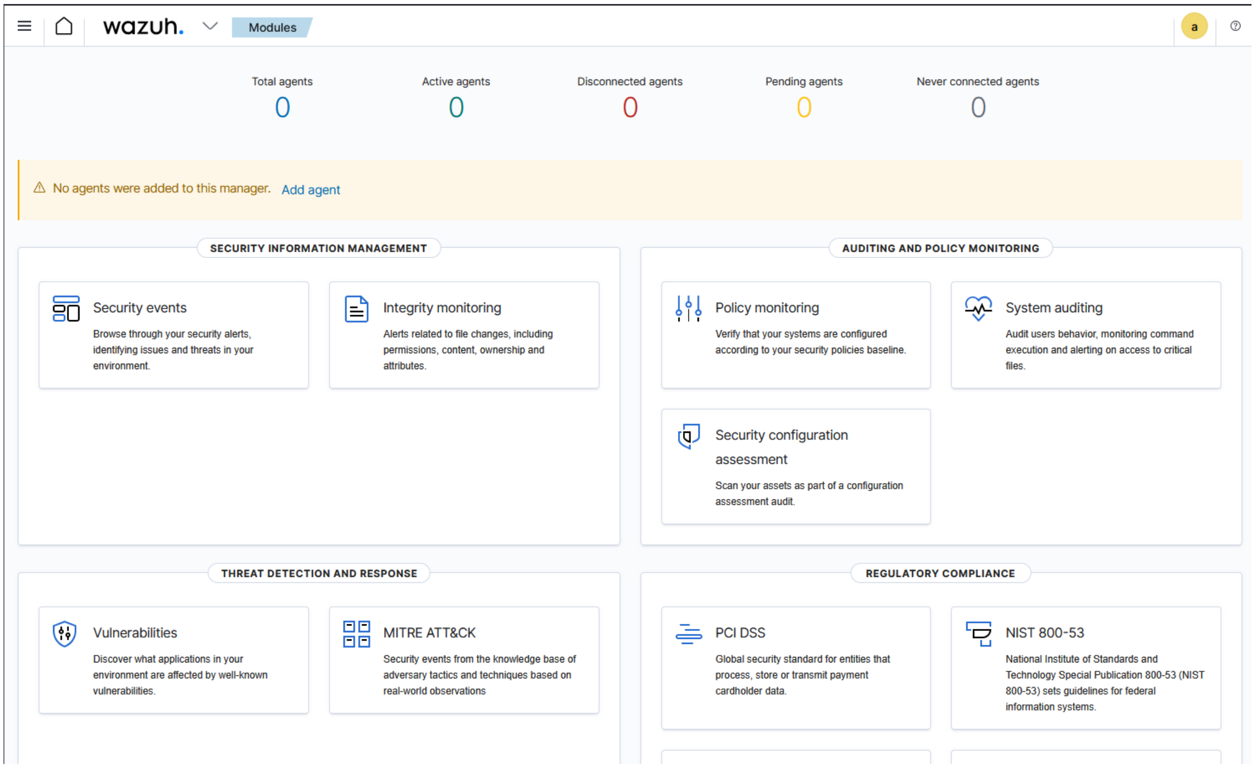
The Wazuh all-in-one deployment method was selected to simplify installation and management.

Components Installed

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

Validation

Deployment was verified by accessing the dashboard through a web browser and confirming all services were operational.



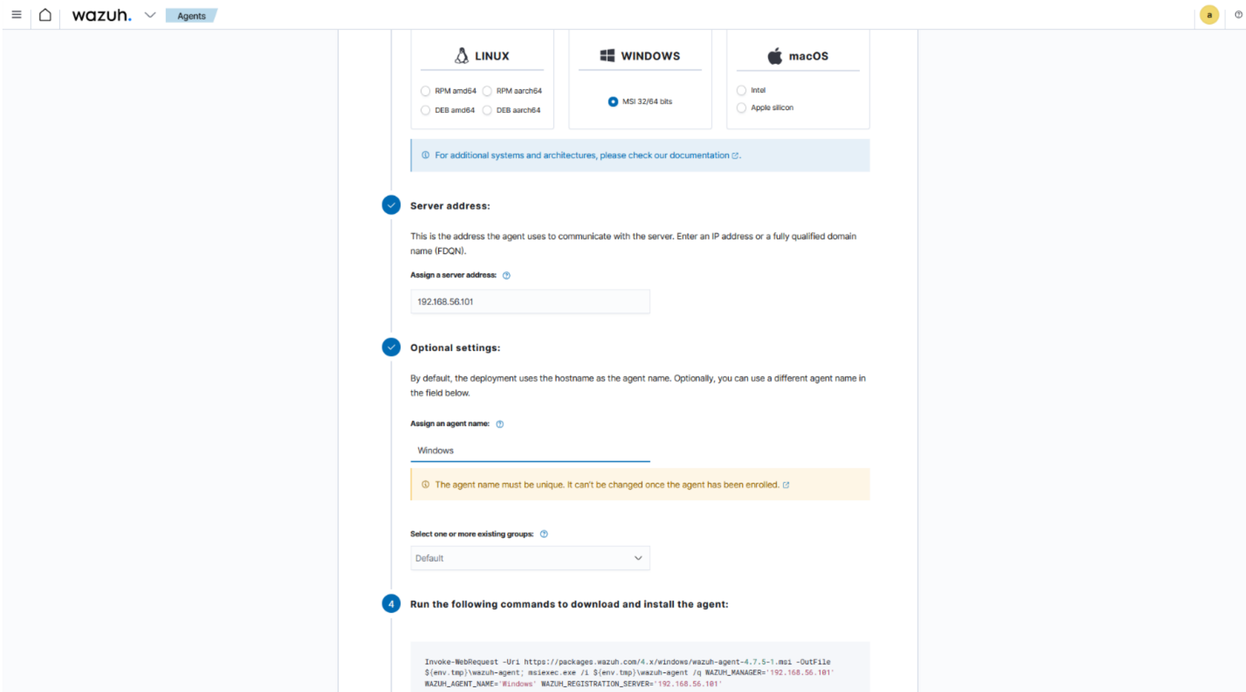
Endpoint Enrollment

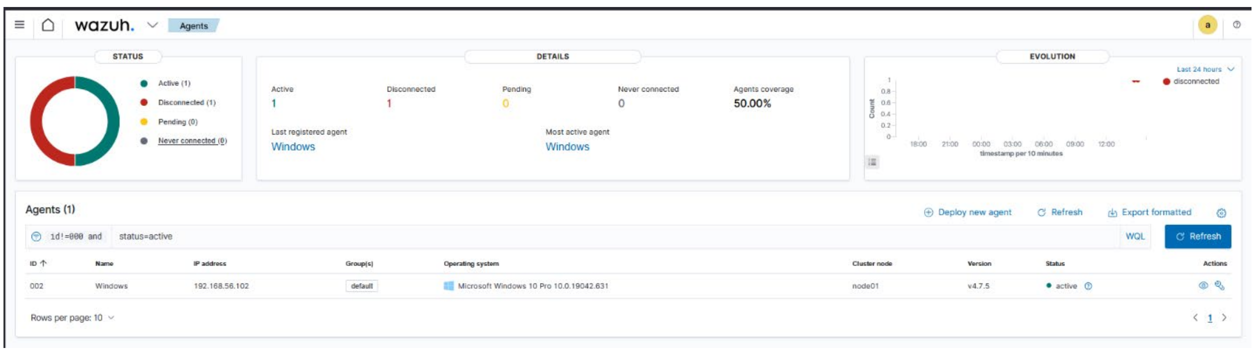
Agent Installation

A Wazuh agent was installed on the Windows endpoint and configured to communicate with the Wazuh Manager.

Verification Steps

- Agent service started successfully
- Endpoint appeared in dashboard inventory
- Telemetry was received





Telemetry Validation

Prior to attack simulation, event collection was verified.

Events Confirmed

- Authentication Events
- Process Events
- Security Logs
- Endpoint Status Information

Successful telemetry collection confirmed proper communication between the endpoint and SIEM infrastructure.

Attack Simulation

Reconnaissance Phase

Nmap was used to identify available services on the Windows endpoint.

```
Nmap scan report for 192.168.56.102
Host is up (0.00043s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
445/tcp  open  microsoft-ds?
3389/tcp open  ms-wbt-server Microsoft Terminal Services
MAC Address: 08:00:27:74:36:A1 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2019|10|XP (91%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_xp::sp3
Aggressive OS guesses: Microsoft Windows Server 2019 (91%), Microsoft Windows 10 1909 (90%), Microsoft Windows XP SP3 (85%), Microsoft Windows XP SP2 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
```

Brute Force Phase

Hydra was used to perform a password-guessing attack against the Windows Administrator account.

```
┌──(andre㉿kali)-[/usr/share/wordlists]
└─$ hydra -l Administrator -P /usr/share/wordlists/rockyou.txt 192.168.56.102 rdp -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes
 (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-28 16:15:40
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking rdp://192.168.56.102:3389/

1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-28 16:15:41
```

Goal

Generate authentication failures and validate SIEM detection capabilities.

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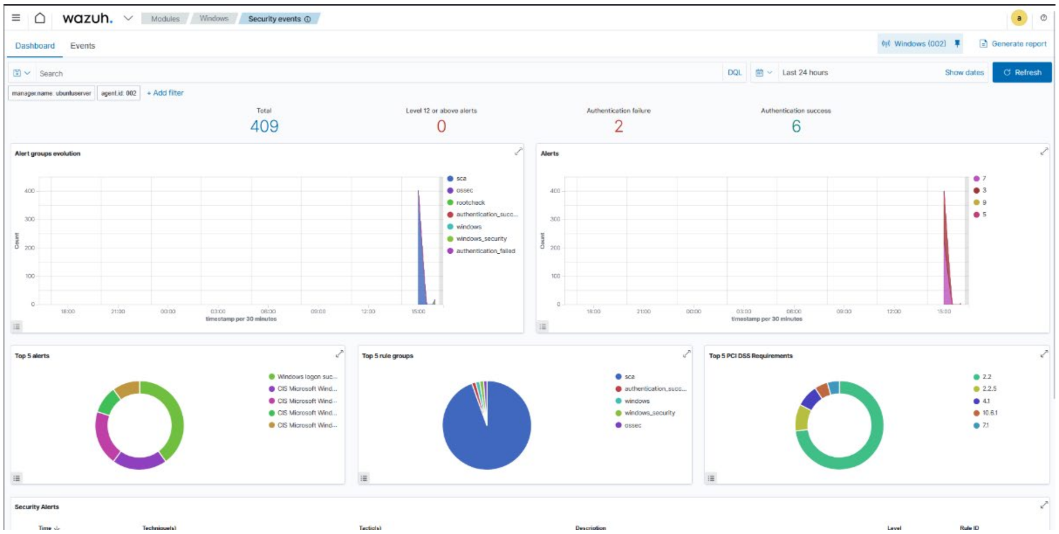
Detection Validation

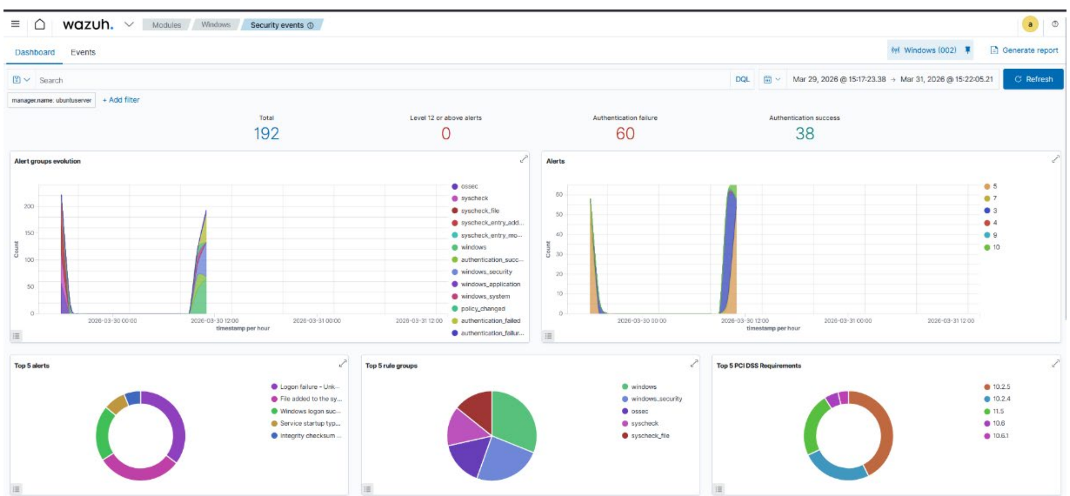
Dashboard Monitoring

During the attack, the Wazuh dashboard reflected increased event activity



Event Collection

Security events were collected and stored successfully.



Event Investigation

Detailed logs provided visibility into attacker actions and authentication attempts.

data.win.system.message	"An account failed to log on.

```
"An account failed to log on.

Subject:
	Security ID:		S-1-0-0
	Account Name:		-
	Account Domain:		-
	Logon ID:		0x0

Logon Type:			3

Account For Which Logon Failed:
	Security ID:		S-1-0-0
	Account Name:		Administrator
	Account Domain:

Failure Information:
	Failure Reason:		Unknown user name or bad password.
	Status:			0xC000006D
	Sub Status:		0xC000006A

Process Information:
	Caller Process ID:	0x0
	Caller Process Name:	-

Network Information:
	Workstation Name:	kali
	Source Network Address:	192.168.56.103
	Source Port:		0

Detailed Authentication Information:
	Logon Process:		NtLmSsp
	Authentication Package:	NTLM
	Transited Services:	-
	Package Name (NTLM only):	-
	Key Length:		0
```

Alert Generation

Security alerts were generated and categorized automatically.

Security Alerts

Time ↓	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 30, 2026 @ 11:24:23.772	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
Mar 30, 2026 @ 11:24:23.740	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
Mar 30, 2026 @ 11:24:23.711	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
Mar 30, 2026 @ 11:24:23.709	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
Mar 30, 2026 @ 11:24:22.538	T1110	Credential Access	Multiple Windows logon failures.	10	60204

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<u>**Challenges Encountered**</u>

Windows Firewall Restrictions

Issue

Kali Linux could not initially communicate with the Windows endpoint.

Resolution

Firewall rules were modified to permit required ICMP and SMB traffic.

Agent Installation Error

Issue

The generated PowerShell installation command contained formatting issues.

Resolution

The installation process was performed manually and validated through service status checks.

<u>**Results**</u>

The lab successfully demonstrated:

- SIEM deployment
- Endpoint monitoring
- Security event collection
- Attack simulation
- Alert generation
- Event investigation
- MITRE ATT&CK mapping